

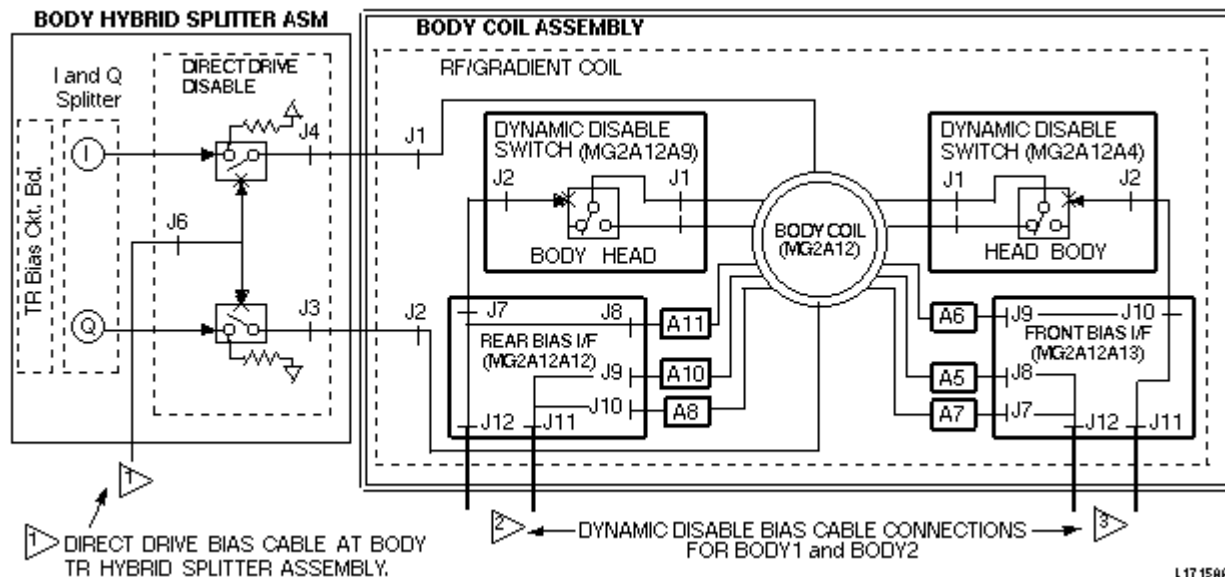
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1 - OVERVIEW

These tests are written for 8.X and Twin Release: RF/PEN1 (with RFSC), RF/PEN2 (with SSM), RF/PDU, and SRF style Cabinets (with SSM).

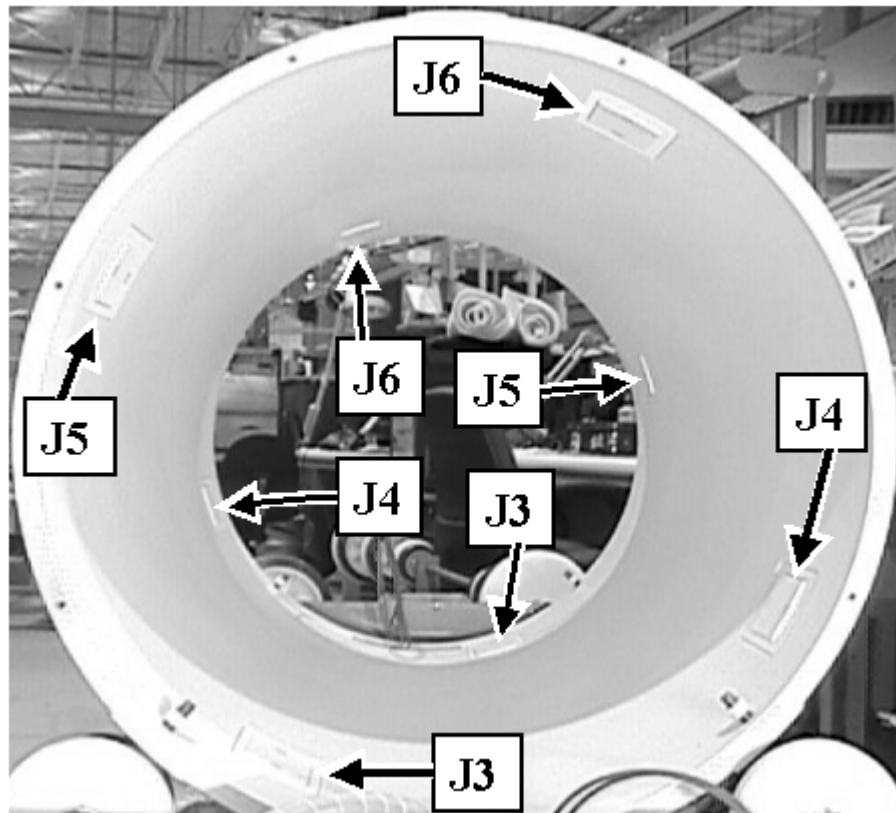
This procedure isolates T/R Switch Circuitry (Head or Body) and Dynamic Disable Circuitry (Direct Drive and Dynamic Disable Switch Circuitry) with faulty PIN diodes (leaky diodes as tested when reverse biased). Illustration 1-1 shows the various bias connections that can be tested (connected vs. disconnected) to isolate a section of switches for Top Level Test (TLT).



PIN DIODE TESTED DURING BODY PROCEDURE (1.0T & 1.5T)
ILLUSTRATION 1-1

1 – OVERVIEW (CONTINUED)

Illustration 1-2 and Table 1-1 show how the dynamic disable switch boards (or in the case of the Twin system, diode boards) correspond to the coaxial drive bias lines. While Illustration 1-2 is that of a Twin RF body coil, the configuration for the LX RF body coil is the same.



REAR VIEW INTO RF COIL SHOWING BOARD POSITIONS CORRESPONDING TO BIAS CABLES
ILLUSTRATION 1-2

TABLE 1-1
DYNAMIC DISABLE BIAS CONNECTIONS

COIL SIDE	BIAS CABLES	DIODE BOARD POSITION
FRONT	J3/J5	J3, J5 (FRONT)
FRONT	J4/J6	J4, J6 (FRONT)
REAR	J3/J5	J3, J5 (REAR)
REAR	J4/J6	J4, J6 (REAR)

TLT will be used to compare noise for the switch circuit tests with bias power connected and disconnected.

Note

It is important to understand that this test does not discuss or isolate a problem due to noise generated by the RF Amplifier (an unlikely and extremely rare occurrence). To isolate this potential problem, the RF Out line should be disconnected (must be a no RF scan).

1-1 Tools Required

Phantoms required (shipped with each system):

- NiCl₂ TLT Head Sphere, 46-265826G6
- Head Loader, 46-287899G1
- Head Coil Tuning Ring, 46-287416G1
- NiCl₂ TLT Body Sphere, 46-265635G6
- Body Loader, 2135652-2

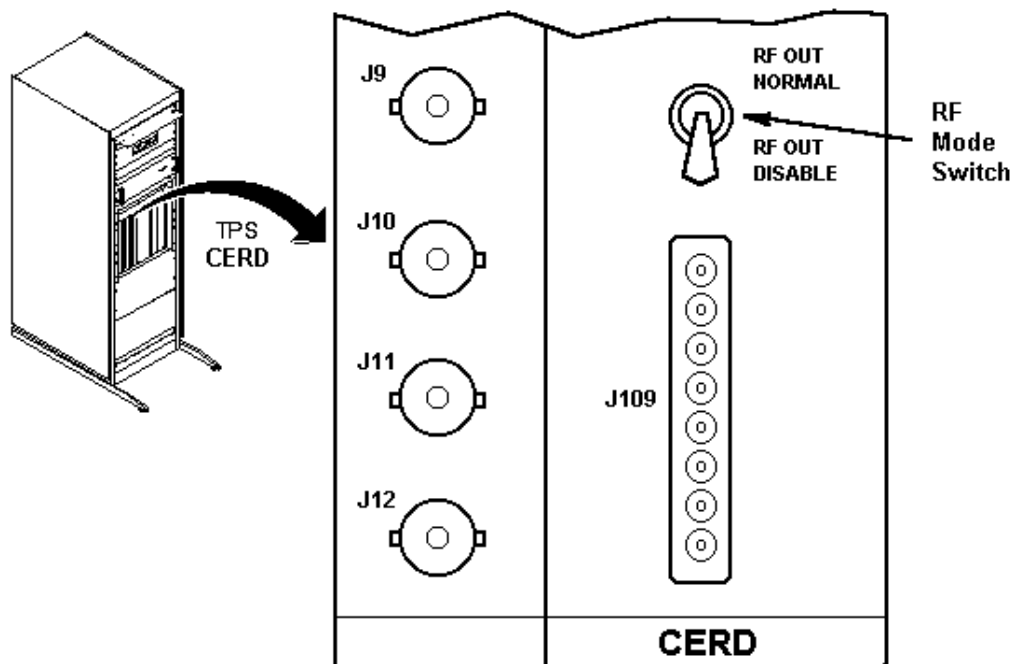


POISON HAZARD! PHANTOM CONTAINS NICKEL, A SUSPECT CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.

2 - HEAD PIN DIODE TEST

2-1 Head Phantom Coil Setup and Test

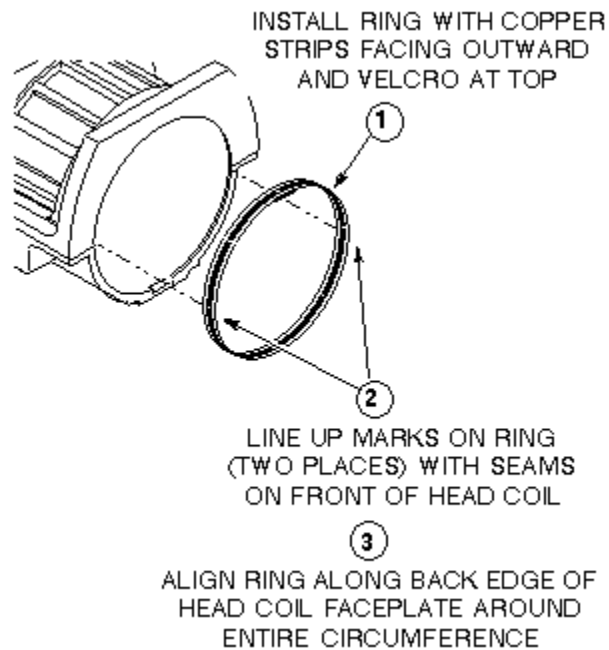
1. Disable the Exciter RF Output, by setting the RF Output mode toggle switch (located on the front panel of the CERD or UCERD), to the RF Out Disable position. (See Illustration 2-1).



RF DISABLED FOR SCAN
ILLUSTRATION 2-1

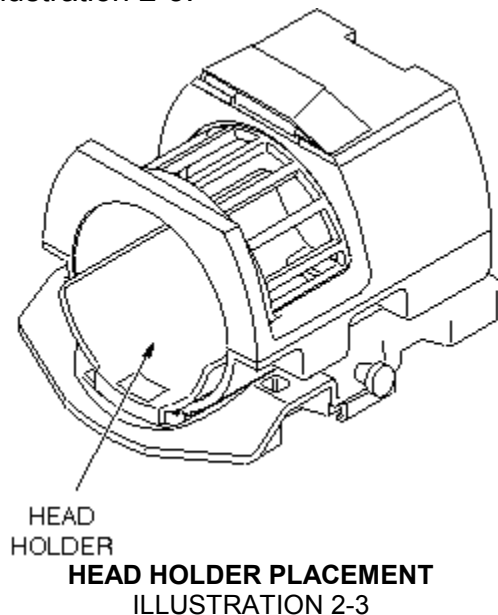
2-1 Head Phantom Coil Setup and Test (Continued)

2. **1.5T ONLY:** Install Tuning Ring in Head Coil with Velcro® connector at top of coil (so that Velcro doesn't rub on head holder when coil top is positioned). See Illustration 2-2.



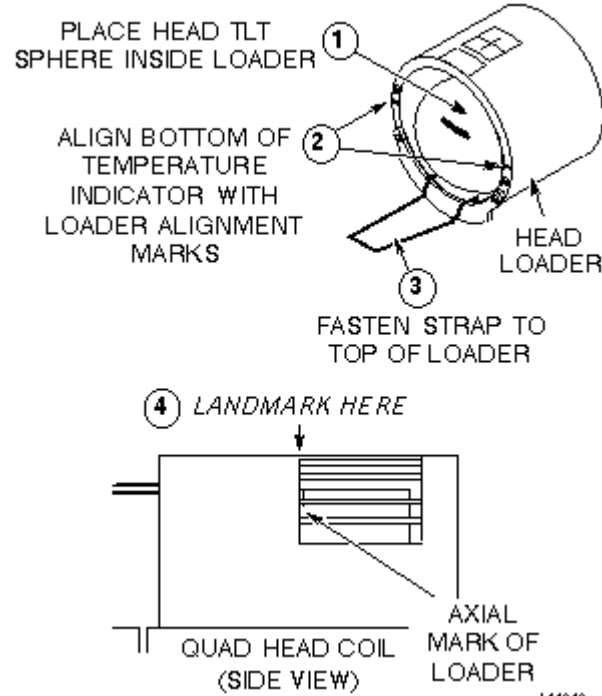
TUNING RING INSTALLATION (1.5T ONLY) ILLUSTRATION 2-2

3. Install head holder. See Illustration 2-3.



2-1 Head Phantom Coil Setup and Test (Continued)

4. Position and landmark TLT head phantom/loader per Illustration 2-4.



HEAD TLT PHANTOM SETUP
ILLUSTRATION 2-4

5. At Operator Workspace, prepare to scan as follows:
- Click on **[New Pt]**, and enter
Id: **geservice**
Name: **pin diode test**
Weight (Lb): **111**
Set Patient Protocols to **Service**.
 - In the Protocol field, type **o.25.1** (o=Other, 1=series number) to load the protocol.
 - [Save Series]**, then **[Prepare to Scan]**.
 - Right click on **[Research Operations]**, select **Setup Params**, and enter settings:
R1= **11**
R2= **15**
TG= **0**
Click on **[Done]**.
 - [Manual Prescan]**.
6. Wait three minutes for preamp warm-up if no TNF is present.
7. **[Scan]**.

Note

Important! Don't start a new TLT scan until the previous TLT analysis is finished or analysis for the new scan may fail to start (only one set of scan data can be analyzed at a time).

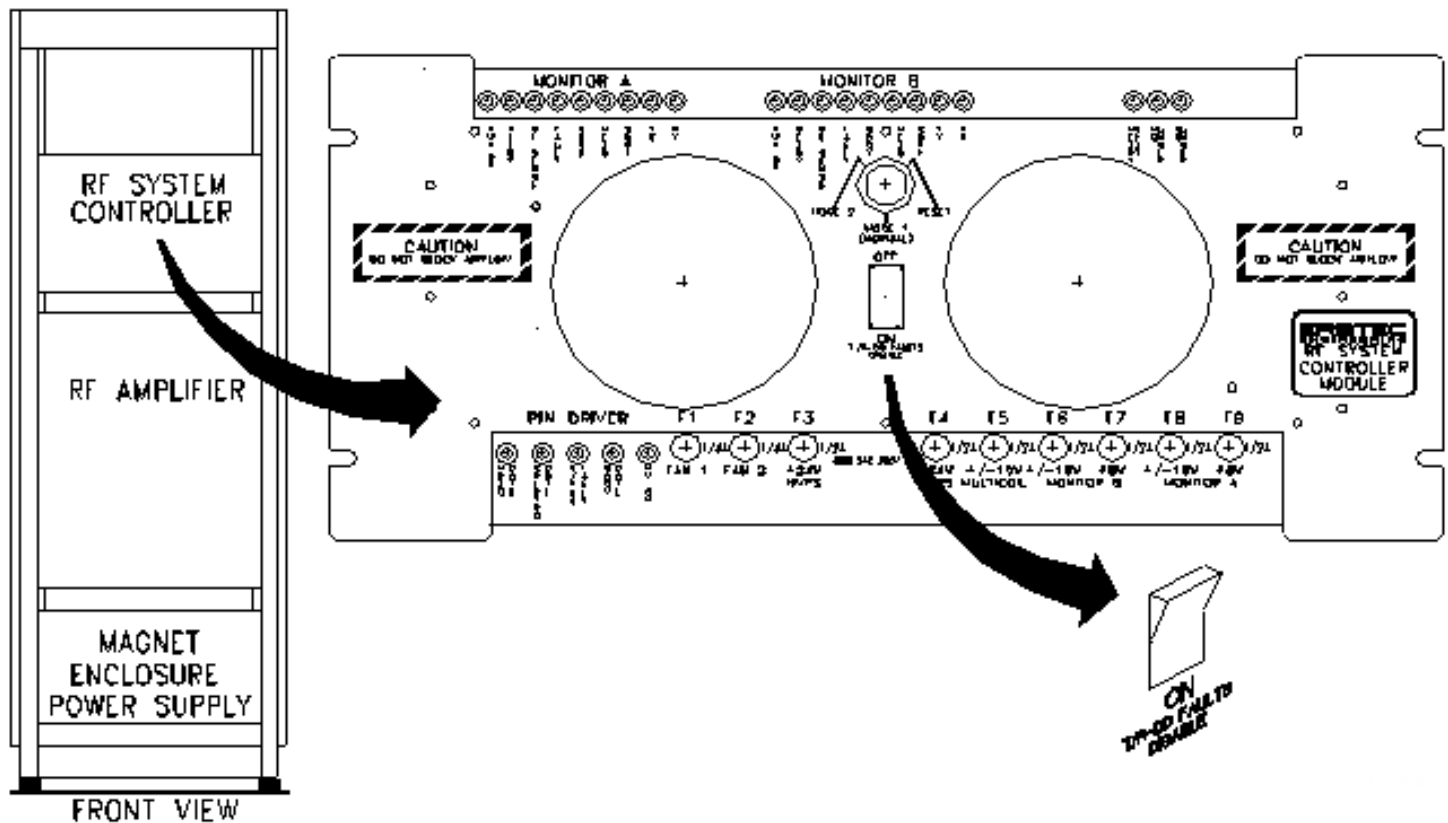
2-1 Head Phantom Coil Setup and Test (Continued)

8. Wait for scan to complete.
9. When scanning and image reconstruction completes, automatic TLT data analysis begins in a window. The analysis provides messages that identify each test as its data is analyzed.

Note

If ASC Analysis option was selected or all tests were run, the system attempts to dial into the ASC for Expert analysis of the data. If successful, results are returned to the site.

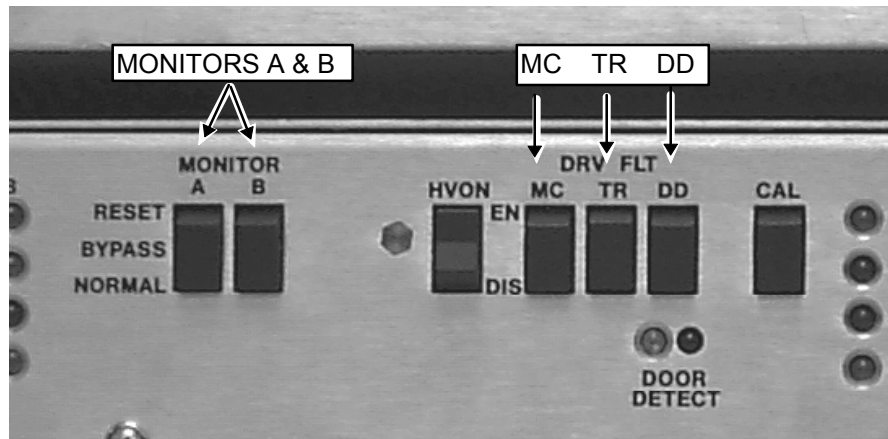
10. Wait for scan to complete.
11. Disconnect J409 (Head T/R Bias) on the rear panel of the RFSC module/SSM.
12. Disable TR Fault detection as follows depending on hardware per Illustration 2-6 or 2-7:
 - A. See Illustration 2-6. PIN Switch Driver Board Fault Detection by setting switch SW2 on the RF System Controller (RFSC) module to ON (faults disabled).



RFSC T/R- DD FAULTS SWITCH TO ON POSITION
ILLUSTRATION 2-6

2-1 Head Phantom Coil Setup and Test (Continued)

- B. See Illustration 2-7. On the front of the SSM place the:
- (three) DRV FLT switches to the DIS (disable faults) position.



SSM FRONT PANEL SWITCHES DISABLED
ILLUSTRATION 2-7



Possible Equipment Damage. Do not select [Manual Prescan], [Auto Prescan], or [Scan] after any Bias lines are disconnected with the RF OUT enabled. The RF chain components may be damaged.

13. Select [Scan].
14. On the Service Desktop, select [Utilities], then [Report Manager] to view TLT results. Compare the noise value for the Head TR Bias cable “disconnected” to the noise value for the Head TR Bias cable “connected” (Refer to Section 4, TLT Results if needed.) Record values in Appendix A, Data Sheet 1.5T or Data Sheet 1.0T.
15. If there is less than a 0.06 difference, the pin diodes are not leaky (normally the noise values should be very close if the results are good). Continue with Section 6- Restoration or Section 3 Body Coil.

Note

If there is more than a difference of 0.06 (1.5T / 1.0T), then a leaky PIN diode is indicated in the Head T/R switch, or the Head preamplifier.

16. Re-connect J409 (Head T/R Bias) on the rear panel of the RFSC module/SSM.
17. Remove Head phantom and loader.
18. Remove Head Coil from magnet bore.

3 - BODY PIN DIODE TEST

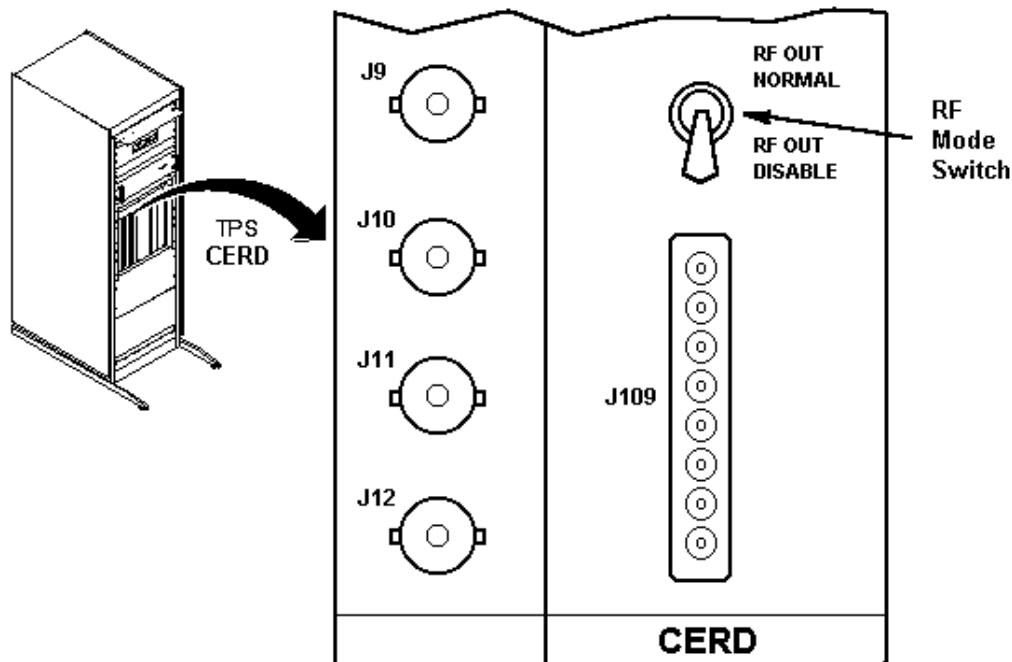
3-1 Body Phantom Coil Setup and Test

1. Remove Head Coil if present.



PROPERTY DAMAGE! TO PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, REMOVE ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

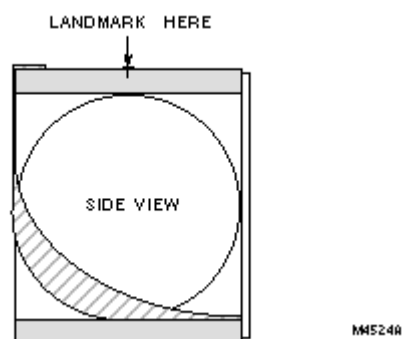
2. Disable the Exciter RF Output, by setting the RF Output mode toggle switch (located on the front panel of the CERD or UCERD), to the RF Out Disable position. (See Illustration 3-1).



RF DISABLED FOR SCAN
ILLUSTRATION 3-1

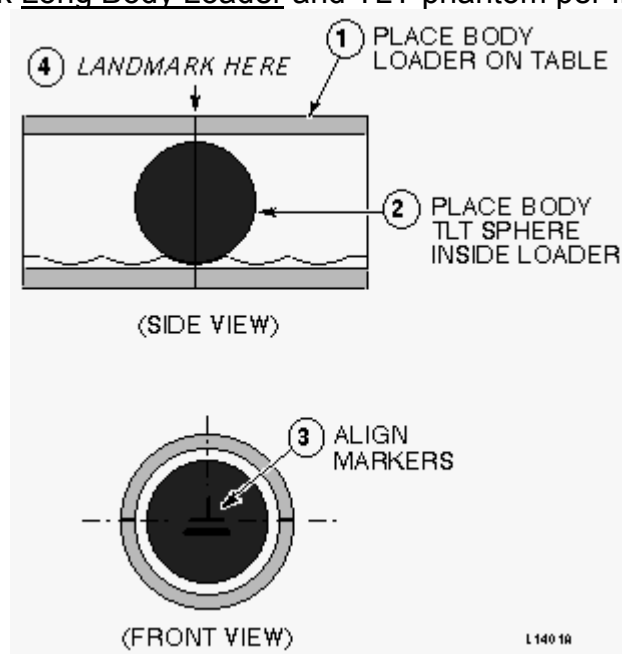
3-1 Body Phantom Coil Setup and Test (Continued)

3. Position the Phantom and Loader, depending on type, per one of the following steps:
A. SPT Body Loader and TLT sphere on the table and landmark per Illustration 3-2.



SPT BODY LOADER AND TLT PHANTOM SETUP
ILLUSTRATION 3-2

- B. Position and landmark Long Body Loader and TLT phantom per Illustration 3-3.



BODY TLT PHANTOM SETUP
ILLUSTRATION 3-3

3-1 Body Phantom Coil Setup and Test (Continued)

4. At operator work space, prepare to scan as follows:
 - a. Click on **[New Pt]**, and enter
Id: **geservice**
Name: **pin diode test**
Weight (Lb): **111**
Set Patient Protocols to **Service**.
 - b. In the Protocol field, type **o.25.2** (o=Other, 2=series number) to load the protocol.
 - c. **[Save Series]**, then **[Prepare to Scan]**.
 - d. Right click on **[Research Operations]**, select **[Setup Params]**, and enter settings:
R1= **11**
R2= **15**
TG= **0**
Click on **[Done]**.
 - e. **[Manual Prescan]**.

Note

On the Patient Position screen, up to 29 characters can be typed in the Series description field as "comments." (Comments are displayed on the Report Header Info screen.)

5. Wait three minutes for preamp warm-up if no TNF is present.
6. **[Scan]**.
7. Wait for scan to complete.

Note

Total time for Body TLT Noise test is ~41 seconds.

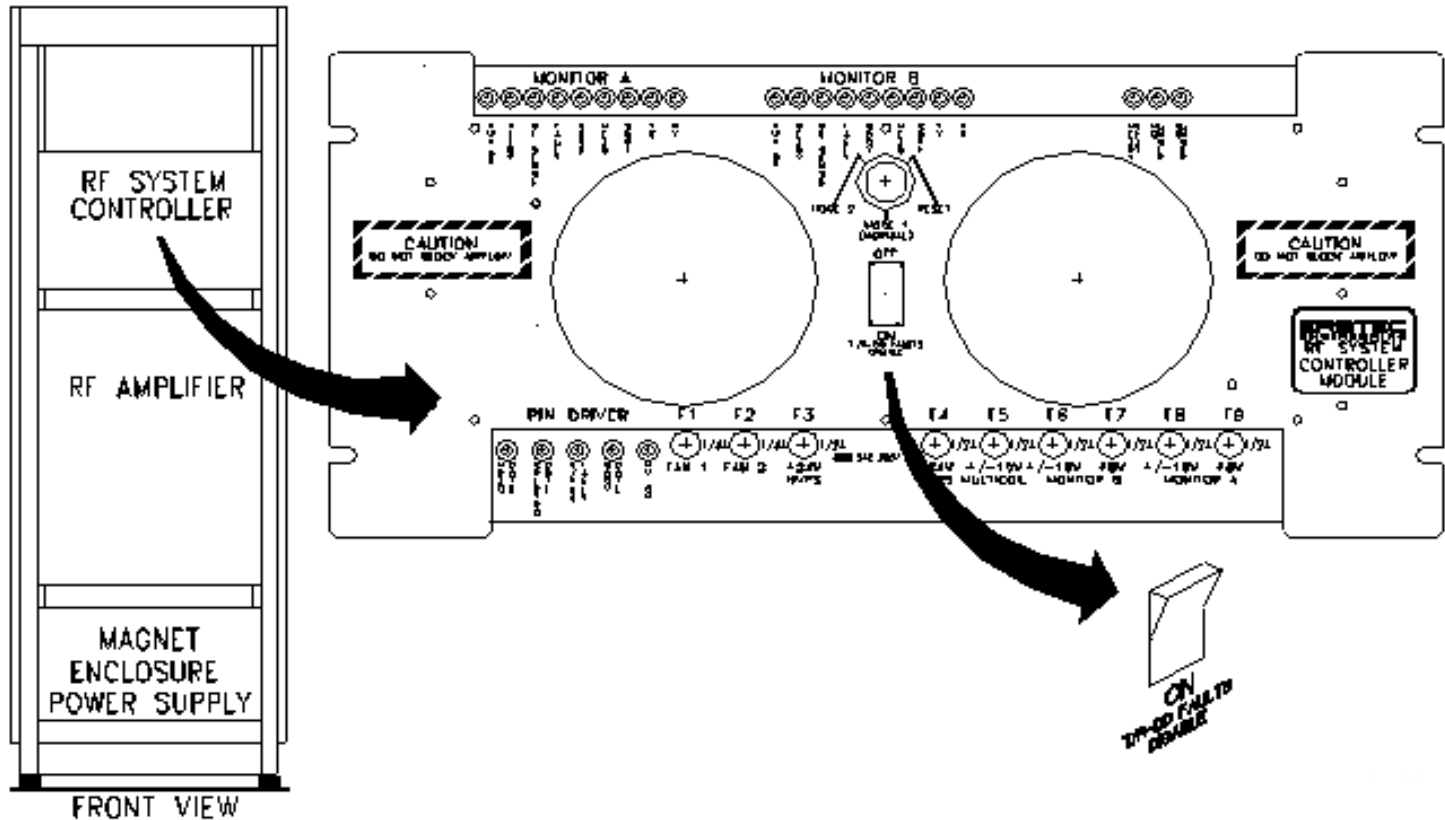
Note

Important! Do not start a new TLT scan until the previous TLT analysis is finished or analysis for the new scan may fail to start (only one set of scan data can be analyzed at a time).

8. Disconnect the following cables:
 - a. Disconnect J408 (Body T/R Bias) on the rear panel of the RF System Controller Module or SSM.
 - b. J403/J42-Body1, J401/J43-Body2, and J402/J44-DD (various Dynamic Disable Bias cables) on the rear panel of the RF System Controller Module/SSM.

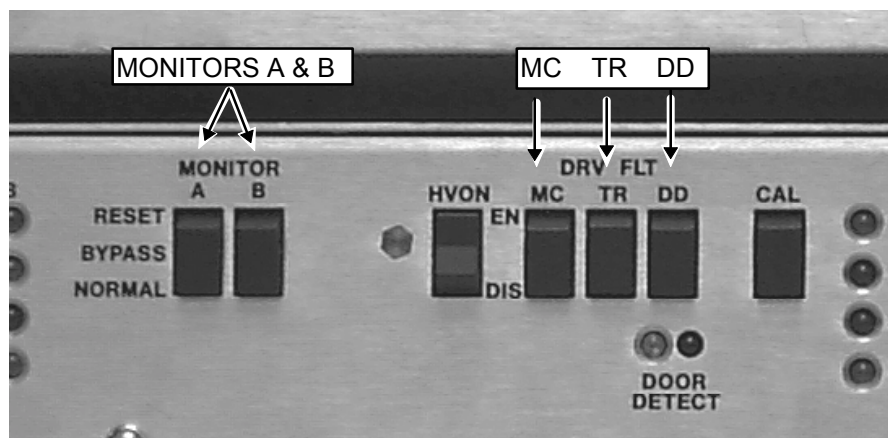
3-1 Body Phantom Coil Setup and Test (Continued)

9. Disable TR Fault detection as follows depending on hardware per Illustration 3-4 or 3-5:
 - A. See Illustration 3-4. PIN Switch Driver Board Fault Detection by setting switch SW2 on the RF System Controller (RFSC) module to ON (faults disabled).



RFSC T/R- DD FAULTS SWITCH TO ON POSITION
ILLUSTRATION 3-4

- B. See Illustration 3-5. On the front of the SSM place the:
 - (three) DRV FLT switches to the DIS (disable faults) position.



SSM FRONT PANEL SWITCHES DISABLED
ILLUSTRATION 3-5

3-1 Body Phantom Coil Setup and Test (Continued)



Possible Equipment Damage. Do not select [Manual Prescan], [Auto Prescan], or [Scan] after any Bias lines are disconnected with the RF OUT enabled. The RF chain components may be damaged.

10. Select **[Scan]**.
11. On the Service Desktop, select **[Utilities]**, then **[Report Manager]** to view TLT results. Compare the noise value for the bias cables disconnected to the noise value for the bias cables connected (Refer to Section 4, TLT Results if needed Record values in Section 7, Data Sheet 1.5T or Data Sheet 1.0T).

If there is less than a 0.06 (**1.5T / 1.0T**) difference, the pin diodes are not leaky. Continue with Section 6- Restoration. If there is more than a difference of 0.06, then a leaky PIN diode is indicated in one of the pin diode circuits.

12. Reconnect ONLY the Body TR Bias line. After the scan compare the “Body TR Bias line reconnect noise value” with the “noise value of all lines disconnected” as performed earlier in this section. View results and determine next step.
13. Reconnect ONLY one of the three Dynamic Disable related pin diode circuit Bias cables at a time. Select **[Scan]**. After each scan, disconnect the bias cable and connect another one. View results and determine next step.
14. If not already done, after all DD related Bias cables have been reconnected one at a time, run the Report tool. For each scan, look for a difference of 0.06 or greater in the noise value between **all** bias cables disconnected and **each** of the bias cables reconnected. This indicates the circuit(s) with leaky diode(s).
15. Re-connect the J408 Body TR Bias cable at the rear of the RFSC/SSM.
16. Re-connect the 3 Dynamic Disable Bias cables at the rear of the RFSC/SSM (J403/J42-Body1, J401/J43-Body2, and J402/J44-DD).
17. Proceed to Section 6, System Restoration.

4 - TLT RESULTS

Top Level Test (TLT) results are viewed (both graphics and text) using the **[Report Manager]** Tool located on the Service Desktop under **[Utilities]**.

See Illustration 4-1 below for an example file name. Refer to Table 4-1 for data file naming convention.

MONTH HOUR SECONDS
17FG4003.TLT
YEAR DAY MINUTES TEST TYPE

* EXAMPLE FILE CREATED: JULY 6, 1991 at 7:40:03

ILLUSTRATION 4-1
EXAMPLE TLT FILE NAME

TABLE 4-1
TABLE 3: DATA FILE NAMING CONVENTION

YEAR	MONTH	DAY		HOUR		MINUTES, SECONDS
0 = 1990	1 = JAN.	1 = 01	H = 17	0 = 00:00	D = 13:00	0 to 59
1 = 1991	2 = FEB.	2 = 02	I = 18	1 = 01:00	E = 14:00	
2 = 1992	3 = MAR.	3 = 03	J = 19	2 = 02:00	F = 15:00	
3 = 1993	4 = APR.	4 = 04	K = 20	3 = 03:00	G = 16:00	
4 = 1994	5 = MAY	5 = 05	L = 21	4 = 04:00	H = 17:00	
5 = 1995	6 = JUN.	6 = 06	M = 22	5 = 05:00	I = 18:00	
6 = 1996	7 = JUL.	7 = 07	N = 23	6 = 06:00	J = 19:00	
7 = 1997	8 = AUG.	8 = 08	O = 24	7 = 07:00	K = 20:00	
8 = 1998	9 = SEP.	9 = 09	P = 25	8 = 08:00	L = 21:00	
9 = 1999	A = OCT.	A = 10	Q = 26	9 = 09:00	M = 22:00	
A = 2000	B = NOV.	B = 11	R = 27	A = 10:00	N = 23:00	
B = 2001	C = DEC.	C = 12	S = 28	B = 11:00	O = 24:00	
:		D = 13	T = 29	C = 12:00		
Z = 2026		E = 14	U = 30			
		F = 15	V = 31			
		G = 16				

4-1 Example TLT Report Screens

The following screens contain example TLT Body test results displayed by the PCReport Tool:

4- TLT RESULTS (CONTINUED)

Scan Header - Screen 1.

SCREEN 1 SCAN HEADER

```

3A193912.TLT/0 Header Info RDF/GRP Revision: 1.0 /43
=====
SITENAME      = BAY 12
USN           = B15B0C0
MLN           = 9999
SRVCONFIG     = 09/27/93 15:28:55
ENCITER       = 000
RECEIVER      = S0/E0/PE0
NMTRFCOIL     = BODY
RCVRFCOIL     = BODY
FREQ          = 63866811 Hz
TIME          = 10/01/93 09:24:00
BASERUN       = 44032
CONFIGCODE    = 001
SOFTREV       = 5.4
NUCLIDE       = 000
HEADERCODE    = 0x00000000
RCVCOILGAIN   = 1.880 R1 = 6 R2 = 14 TG = 135
----- Exam description -----
NO COMMENT
----- Series description -----
DIR15491,SympAnl:IQTS,Sec.
-----
Select (Help, Group #, Next, Files, Quit) [N]:

```

SNR (noise-only scan) - Screen.

SCREEN 2 SNR (NOISE ONLY SCAN)

Record this value →

```

012RRTL.001/12                      AXIAL SNR
=====
Noise      Inoise      Qnoise
1.49       1.50        1.48

      Pk UB      Av UB      Pk B1      Av B1
      Sig        Sig        Sig        Sig
      5102.11    3490.88    8.06      2.34

Plane  Signal      SNR      Area      Hot Pixel
AX      0          0        0          0

      SNR Histograms
-----
Plane  Max      Mean      St Dev
AX      0        0        0

Select (Help, Group #, Previous, Next, Files, Quit) [N]:

```


4- TLT RESULTS (CONTINUED)

Refer to Table 4-2 below for a description of the items on the Header Parameters screen.

TABLE 4-2
REPORT "HEADER INFO" SCREEN PARAMETERS

PARAMETER	LEGAL VALUES	SOURCE
SITENAME	SITE NAME	RAW HEADER
USN	UNIQUE SYSTEM NUMBER (GECARES ISSUED)	CONFIG FILE
MLN	MOBILE LOCATION NUMBER (9999 = NON-MOBIL)	MR CONFIG FILE
SRVCONFIG	DATE/TIME SRV CONFIG FILE LAST CHANGED	SRV CONFIG FILE
EXCITER	(NOT USED AT PRESENT)	(NONE AT PRESENT)
RECEIVER	STARTING RCVR #, ENDING RCVR #; PORT ENABLE #	SCAN Rx; MR CONFIG FILE
XMTRFCOIL	BODY, HEAD	RAW HEADER
RCVRFCOIL	BODY, HEAD, <SURFACE COIL NAME>	RAW HEADER
FREQ	MAGNET FREQUENCY	RAW HEADER
TIME	YY/MM/DD HH:MM:SS	RAW HEADER
BASERUN	BASE RUN NUMBER OF SCAN	RAW HEADER
CONFIGCODE	(NOT USED FOR TLT)	RAW HEADER
SOFTREV	SOFTWARE REVISION	"mrswrev" SCRIPT
NUCLIDE	(NOT USED AT PRESENT)	(NONE AT PRESENT)
HEADERCODE	ERROR # IF PROBLEM CREATING THIS TEST HEADER	CREATED BY TEST ANALYSIS
RCVCOILGAIN	CALIBRATION VALUE (TYP. 1-10)	CONFIG FILE
R1, R2, TG	1-7, 1-30, 0-200	RAW HEADER
Exam Description	UP TO 22 CHARACTERS FROM EXAM DESCRIPTION.	IMAGE HEADER
Series Description	UP TO 29 CHARACTERS FROM SERIES DESCRIPTION.	IMAGE HEADER
"Comments"	UP TO 29 CHARACTERS FROM SERIES DESCRIPTION	IMAGE HEADER

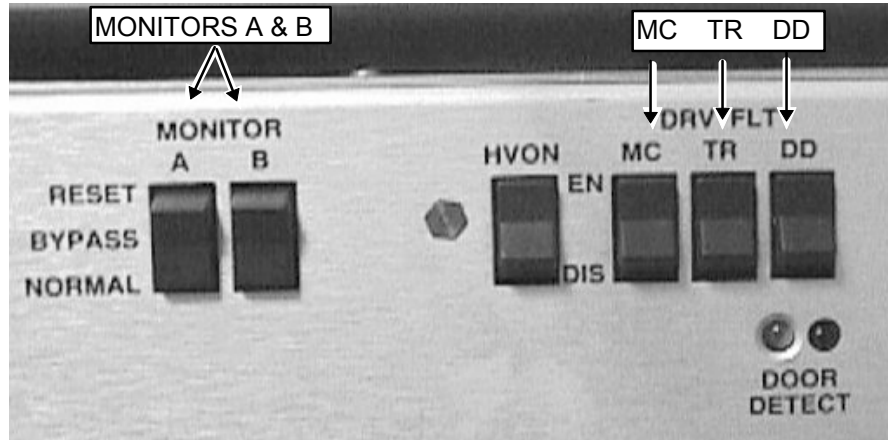
T/R- DD FAULTS SWITCH TO “OFF” POSITION

ILLUSTRATION 6-1

6 - SYSTEM RESTORATION (CONTINUED)

B. Per Illustration 6-2 on the front panel of the SSM place the:

- 2 (two) power MONITOR switches (A and B) to the Normal position.
- 1 (one) HVON switch to the EN (enable faults) position.
- 3 (three) DRV FLT switches to the EN (enable faults) position.



SSM FRONT PANEL SWITCHES ENABLED

ILLUSTRATION 6-2

3. Ensure that all Dynamic Disable Switch Boards are connected.
4. Ensure that all the Dynamic Disable Drive Bias cables (J403/J42-Body1, J401/J43-Body2, J402/J44-Direct Drive) are reconnected at the rear of the RFSC/SSM.
5. Ensure that all the TR Bias cables (J408, J409) are reconnected at the rear of the RFSC/SSM.
6. If the Multi-Nuclear Spectroscopy option is present ensure that the TR Bias cables (J407) is connected at the rear of the RFSC/SSM.
7. Enable the Exciter RF Output, by setting the RF Output mode toggle switch (located on the front panel of the CERD or UCERD), to the RF Out Normal position (see Illustration 1-3).
8. Put all covers back on.
9. Perform one body scan.
10. Perform One Head scan.
11. Delete all unneeded TLT studies in order to maximize image disk space for the customer. (TLT images **are not** deleted automatically in the software.)

7 - DATA SHEETS

1.5T PIN DIODE TEST DATA SHEET

NOISE VALUE	BODY SCAN CONNECTED	BODY SCAN DISCONNECTED	HEAD SCAN CONNECTED	HEAD SCAN DISCONNECTED
		IF FAILING, SEE NOTE 1		IF FAILING, SEE NOTE 2
	SPEC: TLT SNR NOISE DIFFERENCE $\leq .06$		SPEC: TLT SNR NOISE DIFFERENCE $\leq .06$	
NOTE 1: WITH J1 ON REAR OF DYNAMIC DISABLE MODULE CONNECTED, CONNECT BIAS LINES ONE AT A TIME TO DETERMINE CAUSE OF FAILURE. NOTE 2: CHECK HEAD TR SWITCH OR PREAMP.				
CABLES DISCONNECTED				NOISE VALUE
J403/J42	Dynamic Disable Bds 1 & 3			
J403/J44	Dynamic Disable Bds 2 & 4			
J402/J44	Direct Drive			
J401/J43	Dynamic Disable Bds 5 & 7			
J401/J43	Dynamic Disable Bds 6 & 8			
J408	BODY TR & PREAMP			
J409	HEAD TR & PREAMP			
DATE:				

7 - DATA SHEETS (CONTINUED)

1.0T PIN DIODE TEST DATA SHEET

NOISE VALUE	BODY SCAN CONNECTED	BODY SCAN DISCONNECTED	HEAD SCAN CONNECTED	HEAD SCAN DISCONNECTED
		IF FAILING, SEE NOTE 1		IF FAILING, SEE NOTE 2
	SPEC: TLT SNR NOISE DIFFERENCE ≤ 0.04		SPEC: TLT SNR NOISE DIFFERENCE ≤ 0.04	
NOTE 1: WITH J1 ON REAR OF DYNAMIC DISABLE MODULE CONNECTED, CONNECT BIAS LINES ONE AT A TIME TO DETERMINE CAUSE OF FAILURE. NOTE 2: CHECK HEAD TR SWITCH OR PREAMP.				
CABLES DISCONNECTED				NOISE VALUE
J403/J42	Dynamic Disible Brd 1 & 3			
J403/J44	Dynamic Disible Brd 2 & 4			
J402/J44	Direct Drive			
J401/J43	Dynamic Disible Brd 5 & 7			
J401/J43	Dynamic Disible Brd 6 & 8			
J408	BODY TR & PREAMP			
J409	HEAD TR & PREAMP			
DATE:				

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 16, 1998	M. Whitlow	Initial Conversion from Toolbook to Word.
1	Nov 12, 1998	M. Keber	Removed obsolete 8.1 information, added data sheets, misc. style guide cleanup.
2	Oct. 18, 1999	Resa Lambert	Keber added correct proprietary heading to document. Lambert corrected test to include RFSC and SSM Illustrations. Removed redundant (and incorrect) Head and Body Tests. Added SPT loader/phantom to Body test. Clarified test overall.
3	Jan. 10, 2001	Don Thome	Added SRF compatibility.
4	Aug. 16, 2001	Don Thome	Removed references to Quad II coil. Added Illustration and table.